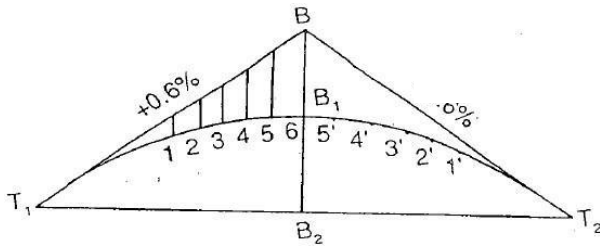


	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch :Civil Engineering Semester :IV Subject Code & Name: BTCVC402, Surveying-II Max Marks: 60 Date: Duration: 3 Hr.																	
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.																	
		(Level/CO)	Marks															
Q. 1	Solve Any Two of the following.			12														
A)	A tacheometer was set up a station C & following reading were obtained on a staff held vertical <table border="1" data-bbox="220 797 1225 1025"> <thead> <tr> <th>Inst. Station</th><th>Staff station</th><th>Vertical angle</th><th>Hair reading(m)</th><th>Remark</th></tr> </thead> <tbody> <tr> <td>C</td><td>BM</td><td>--5°20'</td><td>1.50,1.800,2.450</td><td rowspan="2">RL of BM.=750.50m</td></tr> <tr> <td>C</td><td>D</td><td>+8°12'</td><td>0.750,1.500,2.250</td></tr> </tbody> </table> Calculate the horizontal distance CD & RL of D, when the constants of instruments.		Inst. Station	Staff station	Vertical angle	Hair reading(m)	Remark	C	BM	--5°20'	1.50,1.800,2.450	RL of BM.=750.50m	C	D	+8°12'	0.750,1.500,2.250	CO1	06
Inst. Station	Staff station	Vertical angle	Hair reading(m)	Remark														
C	BM	--5°20'	1.50,1.800,2.450	RL of BM.=750.50m														
C	D	+8°12'	0.750,1.500,2.250															
B)	Explain Anallatic lens in detail with neat labeled diagram.		CO1	06														
C)	Explain the theory of Stadia Tacheometry.		CO2	06														
Q.2	Solve Any Two of the following.			12														
A)	Define and Explain i) Signals ii) Satellite Station iii) Spherical triangle.		CO2	06														
B)	Define Triangulation and Explain Classification of Triangulation System.		CO2	06														
C)	Define Base line, List down base measuring equipments and write down points to be considered while selecting the base line		CO2	06														
Q. 3	Solve Any One of the following.			12														
A)	The meridian altitude of the Sun's lower limb was observed to be 41°12'26" at a place in longitude 72°20'45" W to determine the latitude of the place. The Sun was to the south of the zenith. The declination of the sun at GAN on the day of observation was 19°38'52" N, increasing 7.46" per hour and in semi-diameter 16°14'24". Determine the Latitude of the place.		CO3	12														
B)	Define i) Vertical circle ii) The Prime Vertical iii) The Longitude iv) The Latitude v) The Declination vi) Hour Circle		CO3	12														

C)	Explain method of determination of latitude by meridian altitude of Sun or Star with neat labeled diagram	CO3	12
Q.4	Solve Any Two of the following. \		12
A)	Define Combined Curve. Explain elements of Combined curve with neat labeled diagram.	CO4	06
B)	What is Shift? Prove that a transition curve bisects a shift and that a shift bisects a Transition Curve.	CO4	06
C)	Calculate the RL of the various pegs on a vertical curve connecting two grades of +0.6% and -0.6%. The chainage and the RL of intersection point are 550 and 325.50 m respectively. The rate of change of grade is 0.1% per 30 m. 	CO4	06
Q. 5	Solve Any One of the following.		12
A)	a) A line AB measures 15cm on a photograph taken with a camera having focal length of 21.5cm. The same line measures 5 cm on a map drawn to a scale of 1/45000. If the average altitude is 450 m, Calculate the flying height of the aircraft. b) A line 3350 m long lying at an elevation of 500m measure 10.50 cm on a vertical photograph. The focal length of the camera is 10cm. If the elevation of a point is 1300m. Calculate the scale of photograph.	CO4 CO4	06
B)	a) Write down comparison between Map and Aerial Photograph b) Define: i) Forward overlap ii) Side overlap iii) Relief Displacement	CO5 CO5	06
C)	a) Write down applications of GPS and GIS. b) Explain key component of GIS	CO6 CO6	06
	*** End ***		

The grid and the borders of the table will be hidden before final printing.